What Clinical Researchers Can Borrow from Federated Learning

A multicentre clinical AI research lens

Use this before the first protocol meeting. The point is not to force federated learning into every project; it is to make collaboration design, heterogeneity, privacy, evaluation, and governance explicit.

1 Collaboration design

Name what moves: data, updates, metrics, aggregates, or conclusions. Justify why that movement is needed.

2 Institution as a variable

Treat site, workflow, scanner, assay, referral, and annotation differences as measurable scientific context.

3 Missingness and modality

Make the modality-by-site matrix early. Decide what common core is possible and what remains local.

4 Site-level evaluation

Predefine global, per-site, worst-site, subgroup, calibration, uncertainty, abstention, unseen-site, and drift criteria.

5 Privacy as information flow

Write who can see what, what can be inferred, what is logged, what is released, and who responds.

6 Governance and maintenance

Assign clinical, data, security, ethics, contract, authorship, monitoring, withdrawal, and update responsibilities.

Protocol sentence: The study will be considered unsuccessful if any site, subgroup, or calibration stratum falls below the predefined clinical threshold.